

MA 442 - Quiz

April 29

Name: _____ BUID: _____

Question 1. Suppose that a 3×3 matrix A of rank 2 satisfies

$$A^3 = A \tag{1}$$

- (a) Prove that A is diagonalizable.
- (b) Enumerate **all** of the possible characteristic polynomials of A .

Solution. Let $p(t) = t^3 - t$. Then A satisfies the polynomial equation $p(A) = 0$. This means that if λ is any eigenvalue of A that $p(\lambda) = 0$ as well. Since $t^3 - t = t(t - 1)(t + 1)$ this means that the possible eigenvalues of A are $0, \pm 1$.

- (a) Let $v \in \mathbb{R}^3$ and define the following vectors

$$\begin{aligned} v_0 &= v - A^2v \\ v_1 &= \frac{1}{2}(A^2v - Av) \\ v_{-1} &= \frac{1}{2}(A^2v + Av) \end{aligned}$$

Then v_0, v_1, v_{-1} are eigenvectors with eigenvalues $0, 1, -1$ respectively. Also $v = v_0 + v_1 + v_{-1}$. In other words, any vector can be decomposed uniquely as a sum of eigenvectors for A . Therefore, A is diagonalizable.

- (b) Since A is of rank 2 this means that

$$\dim \ker A = \dim E_0 = 3 - 2 = 1. \tag{2}$$

Since A is diagonalizable, this means that the multiplicity of the eigenvector 0 is also 1. There are three possibilities for $\chi_A(t)$:

$$-t(t - 1)^2, \quad -t(t^2 - 1), \quad -t(t + 1)^2. \tag{3}$$